NEXORA · AI & DATA SCIENCE

Feasibility Study

DAMI

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The right blood unit, to the right hospital, before it's requested... and by drone when there's no time

A preliminary feasibility study covering market, technical, operational, legal and 3-year financial feasibility, with scenarios, sensitivity and a recommendation.

Team	**NEXORA**
Field	Health · Artificial Intelligence & Data Science · Drones
Competition	Jordan Students Pitch Competition — GEW Jordan
University / Supervisor	.....
Team members	.....

Because blood that arrives late... doesn't arrive.

1 Summary of findings

Initial investment	Funding needed	Year-3 revenue	Year-3 operating result	Year-3 gross margin
JOD 58,000	JOD 89,900	JOD 325,000	JOD 85,250	72%

Dimension	Rating	Why
Market	✅ High	Critical health need, clear customer (hospitals and the blood bank), no direct local competitor per our preliminary research.
Technical	✅ High	Mature technologies (forecasting, optimization, medical drones proven worldwide); the prototype works.
Operational	⚠️ Medium	Needs an operations center, licensed pilots and cold-chain management.
Legal	⚠️ Biggest challenge	Flight permits, security approvals and health-data protection.
Financial	⚠️ Long-term	Year 3 is positive; break-even after year 3 in the base case and month 27 in the optimistic case.

2 Market study

Blood banks face a hard equation: shortages of certain blood types when they are needed, while units expire elsewhere. In major accidents or shortage crises, hospitals need urgent units from the blood bank or another hospital, but vehicles get stuck in traffic or delayed on mountain and remote roads, and in severe bleeding every minute counts.

35–42 days
Shelf life of red blood cells stored at 2–6°C

5–7 days
Shelf life of platelets only, so some are wasted unless distributed smartly

O negative
The type requested first in severe bleeding, before the patient's type is known

Since 2016
Drone blood delivery operates in Rwanda, then Ghana and other countries (Zipline)

Target segments

- Private hospitals (beachhead)
- Ministry of Health hospitals and Royal Medical Services
- National Blood Bank
- Red Crescent and Civil Defense
- Donors (free users)

Growth assumptions

Assumption: 3 private hospitals in year 1, then 10, then 20, with drone flights starting in year 2 after licensing.

Competition and alternatives

Alternative	What it offers	Where it falls short
Status quo	Phone calls between blood banks, transport by car, general donation drives	Slow in emergencies, expiry waste, no type targeting

Blood bank management systems	Stock records inside one hospital	No forecasting, no national view, no delivery
Zipline (global)	Drone delivery in Africa and the US	Not present in Jordan per our preliminary research; focused on delivery only
DAMI	**Forecasting + balancing + donors + drone in one platform**	**Built for Jordan; starts with simulation and phases into licensed operations**

3 Technical feasibility

Forecasting & balancing

A model forecasts 14-day demand for each type at each hospital, warns before shortages, and proposes moving units close to expiry to a hospital that will use them.

Smart donor call-up

Selects eligible donors (last donation 90+ days ago) who are nearby and most likely to respond, and invites them for exactly the type that's short.

Blood ambulance drone

In emergencies the system picks the nearest source holding the type, compares drone vs car time, and launches a drone with a smart cooling box and live temperature and position tracking.

Technology stack

Front end	Web dashboard + mobile donor app (React / Flutter)
Back end	Python (FastAPI) + PostgreSQL + real-time streaming over WebSocket
AI	Time-series models (Prophet / LightGBM) for forecasting, and mathematical optimization for source selection and fleet dispatch
Drone	Integration via MAVLink with the ground station, plus restricted-zone maps
Cooling sensors	IoT temperature loggers reporting every second over cellular
Hosting	Cloud inside Jordan or the health authority's servers, per data-protection requirements

Drone and cooling specification

Type	Multirotor drone for the pilot, then fixed-wing VTOL for longer distances
Payload	2–4 kg: 4–6 blood units with the cooled box
Range & speed	15–40 km for urban pilots, at 60–90 km/h
Cooling box	Thermal insulation and phase-change material holding 2–6 °C, with a temperature logger and live alarm
Connectivity	4G/5G cellular with backup link, streaming position, temperature and battery every second
Safety	Emergency parachute, automatic return on link loss, and geofencing
Delivery	Landing on a dedicated pad on the hospital roof or a small-parachute drop in a safe zone
Operations	A licensed pilot monitors from the operations center, on pre-approved routes

4 Operational feasibility

1. Emergency request
The hospital requests 6 units of O negative in

2. Source selection
The system checks live stock and picks the blood bank or the

3. Launch decision
Comparison: car 34 min at peak vs drone 9 min; the operator approves with one tap.

4. Packing & flight
Units packed in a 2–6 °C cooled box, on a safe route that avoids restricted zones.

5. Delivery & restock
Landing on the hospital pad, receipt and flight temperature

the app with “critical” priority.	nearest hospital holding the type.			confirmed, and donors called to restock.

- A small 24/7 operations center with an operator and a licensed pilot on every shift.
- Routine drone maintenance, box inspection and monthly temperature-sensor calibration.
- Fallback: if flying is not possible (weather, fault), the request automatically switches to road transport from the nearest source.
- Training blood bank staff to pack boxes and receive shipments.

5 Legal and regulatory feasibility

Authority / requirement	What we need
Civil Aviation Regulatory Commission	Drone operating permit, approval of routes and licensed pilot, starting with a pilot route.
Relevant security authorities	Approvals for flights over residential areas and near sensitive sites.
Ministry of Health & National Blood Bank	Partnership agreement and compliance with transfusion and cold-chain standards.
Data protection	Privacy policy, donor consent and compliance with the Personal Data Protection Law.
Insurance	Civil liability insurance on the fleet and shipments.
Company registration	Register a startup and protect the name and trademark.

6 Financial feasibility

Initial investment (CAPEX)

Item	Qty	Unit cost (est.)	Total
Medical multirotor drone, 3–5 kg payload	2	JOD 11,000	JOD 22,000
Smart cooling box with phase-change material and temperature logger	4	JOD 750	JOD 3,000
Spare batteries and fast charger	1	JOD 2,000	JOD 2,000
Landing pad and ground control station	1	JOD 3,000	JOD 3,000
Software platform and forecasting model (development)	1	JOD 15,000	JOD 15,000
Donor app (development)	1	JOD 8,000	JOD 8,000
Cybersecurity and compliance	1	JOD 5,000	JOD 5,000
Total			**JOD 58,000**

Operating expenses (OPEX)

Item	Y1	Y2	Y3
Team salaries	JOD 30,000	JOD 57,000	JOD 90,000
Technology & servers	JOD 5,000	JOD 9,500	JOD 15,000
Marketing & sales	JOD 7,500	JOD 14,250	JOD 22,500
Operations & admin	JOD 7,500	JOD 14,250	JOD 22,500
Total	**JOD 50,000**	**JOD 95,000**	**JOD 150,000**

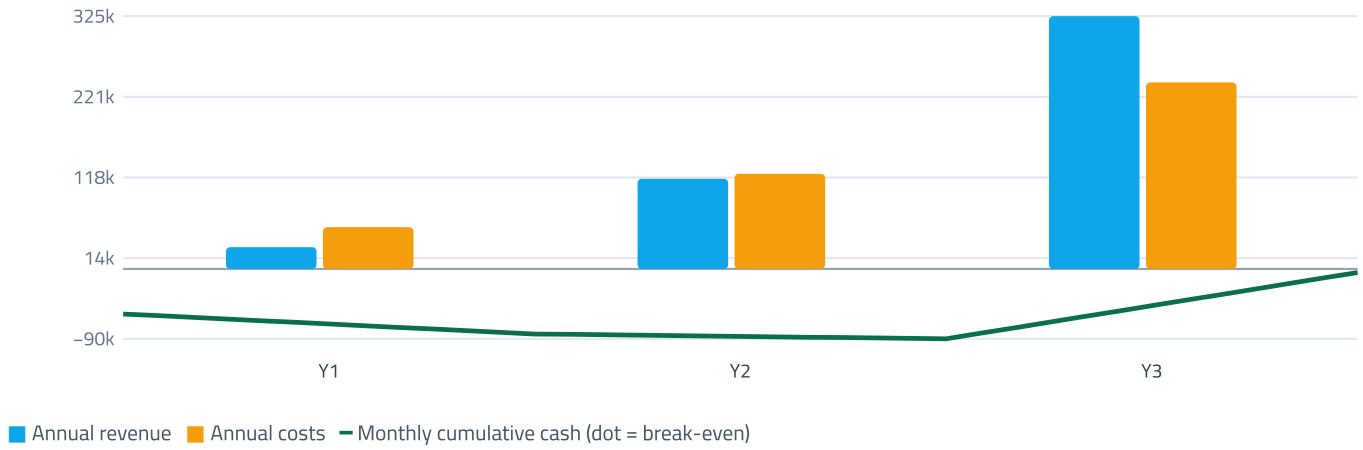
Revenue streams

Stream	Unit	Price	Y1	Y2	Y3	Y3 revenue
Hospital subscription	hospital/year	JOD 6,000	3	10	20	JOD 120,000
Emergency drone delivery	flight	JOD 90	0	400	1,500	JOD 135,000
National dashboard contract	annual contract	JOD 40,000	0	0	1	JOD 40,000
Donor-app sponsorship	sponsor/year	JOD 10,000	1	2	3	JOD 30,000

Projected income statement

Item	Year 1	Year 2	Year 3
Revenue	JOD 28,000	JOD 116,000	JOD 325,000
Direct costs	JOD 3,700	JOD 27,200	JOD 89,750
Gross profit	JOD 24,300	JOD 88,800	JOD 235,250
Operating expenses	JOD 50,000	JOD 95,000	JOD 150,000
Operating result	**–JOD 25,700**	**–JOD 6,200**	**JOD 85,250**
Cumulative cash after investment	**–JOD 83,700**	**–JOD 89,900**	**–JOD 4,650**

36-month cash flow



Unit economics

Metric	Value
Emergency flight price	JOD 90
Direct cost per flight (battery, maintenance, pilot time, insurance)	JOD 40.50
Contribution margin per flight	JOD 49.50 (55%)
Annual hospital subscription / direct cost	JOD 6,000 / JOD 900
Year-3 flights (assumption)	1,500 ≈ 4.1 per day

Sensitivity analysis

Change	Y3 result	Y3 cumulative cash	Break-even
Price −20%	JOD 38,200	−JOD 74,320	After year 3
Price +20%	JOD 132,300	JOD 65,020	Month 31
Volume −20%	JOD 38,200	−JOD 74,320	After year 3
Volume +20%	JOD 132,300	JOD 65,020	Month 31
Costs +20%	JOD 55,250	−JOD 63,650	After year 3
Costs −20%	JOD 115,250	JOD 54,350	Month 31

Scenarios

Scenario	Volume	Costs	Y3 revenue	Y3 result	Funding needed	Break-even
Pessimistic	×0.6	×1.1	JOD 195,000	−JOD 23,850	JOD 173,490	After year 3
Base	×1.0	×1.0	JOD 325,000	JOD 85,250	JOD 89,900	After year 3
Optimistic	×1.4	×0.95	JOD 455,000	JOD 186,850	JOD 71,480	Month 27
Base + grant for pilot fleet	**×1**	**×1**	**JOD 325,000**	**JOD 85,250**	**JOD 59,900**	**Month 33**

7 Risks

Risk	Mitigation
Sensitivity of health data	Formal partnership, encryption and consent

Licensing drone flights in Jordanian airspace	Start with simulation, then a pilot route approved by the Civil Aviation Regulatory Commission and security authorities
Maintaining the cold chain	Medically certified boxes with live temperature monitoring and alarms
Slow government contracting	Start with private hospitals and corporate sponsorship

8 Recommendation

The project is technically feasible with high health impact, and its financial feasibility is long-term. Recommendation: proceed in phases — software first for early revenue, fund the pilot fleet through a grant or health partner, and do not launch commercial flights before permits are complete.

Assumption validation plan

- Interview 10–15 blood bank and private hospital managers to test willingness to pay JOD 6,000 per year.
- Collect real data on expiry waste rates and current emergency transport times.
- Get real quotes for two drones and certified cooling boxes.
- Hold an introductory meeting with the Civil Aviation Regulatory Commission to understand the pilot-permit path.

Financial figures are preliminary planning assumptions set by the team, not market research or supplier quotes, and must be validated before any decision. “Not available in Jordan” means no equivalent was found in preliminary research. The simulator uses synthetic data and schematic locations.